



# CINESCOPE

*What makes a movie successful?*

DATA VISUALIZATION • COM-480 • EPFL 2025

Milestone 2

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## Project Goal

CineScope explores a single overarching question: **what drives financial and critical success in cinema, and how have those relationships shifted across decades?** We treat financial and critical success as two parallel dimensions and examine where they align or diverge. Rather than presenting static conclusions, we build a coordinated interactive system that lets users test their own hypotheses.

Following Milestone 1 feedback, we have added a dedicated view for franchise and sequel analysis, examining whether follow-up films reliably outperform their originals.

**CineScope is designed as a coordinated multiple-view system.** Selecting a genre in the Sankey flow diagram filters the main bubble explorer, while clicking a film in the sequel view highlights its position in the budget-revenue space. A persistent search bar and genre/decade controls remain active across all views, allowing users to pursue one hypothesis across multiple perspectives simultaneously.

## Visualizations

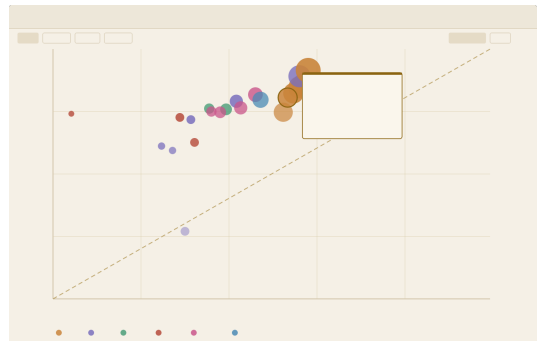
### View 01 — The Dossier Board

Core MVP

A **D3 force-layout beeswarm** where every film is a physics-colliding bubble. Color encodes genre, size encodes budget. The user lands on a clustered galaxy of films and can toggle between three states with animated transitions: (1) genre clusters where bubbles fly apart by category, (2) scatter-plot coordinates mapping budget on x and revenue on y, and (3) a rating vs ROI view for a different definition of success. Hovering any bubble opens a film dossier card with title, director, cast, ROI badge and rating.

**Key interactions:** genre filter, decade slider, lasso selection to isolate outliers, quadrant annotations (blockbusters, cult losses, micro-budget breakouts), toggle between revenue and ROI on y-axis.

**Tools:** D3.js v7 force simulation, D3 brush, CSS transitions · Lectures 4, 5, 6



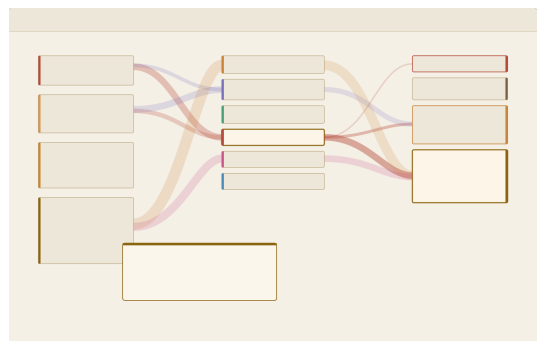
### View 02 — The Flow

Core MVP

An **alluvial / Sankey diagram** mapping the flow of films from budget tier (Micro, Low, Mid, Blockbuster) through genre to financial outcome (Flop, Break-Even, Hit, Megahit). This directly visualizes the core research question in one chart: does a given budget and genre combination reliably lead to profit? Hovering a genre node highlights all paths flowing through it, instantly revealing patterns such as most Horror films starting in the low-budget tier but flowing into the Hit outcome tier.

**Key interactions:** hover to highlight paths, click genre to filter View 01 bubbles, toggle metric between revenue and ROI outcomes.

**Tools:** D3.js v7 Sankey plugin · Lectures 4, 5



### View 03 — The Profitability Matrix

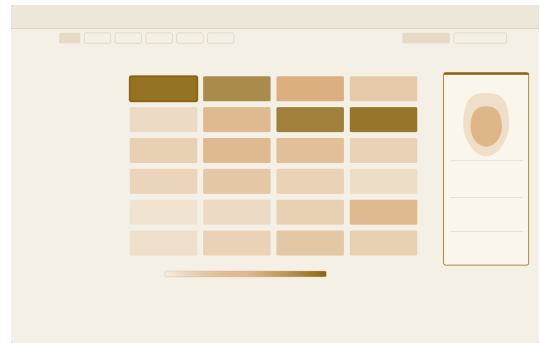
Core MVP

A **genre × budget-tier heatmap** where cell color encodes median ROI or probability of turning a profit. Rows are genres, columns are budget bins from micro to blockbuster. Selecting a decade via a top slider updates the entire matrix with a smooth transition, revealing how profitability patterns have shifted over time. Clicking any cell zooms into a detail panel showing the ROI distribution for that exact genre and budget combination as a violin plot alongside the full-dataset baseline.

This directly answers the project question: given a genre, decade, and budget, what is the likely return?

**Key interactions:** decade slider, cell click to drill down, toggle between median ROI and probability of profit, comparison against full dataset baseline.

**Tools:** D3.js v7 heatmap, violin plot · Lectures 4, 5, 7



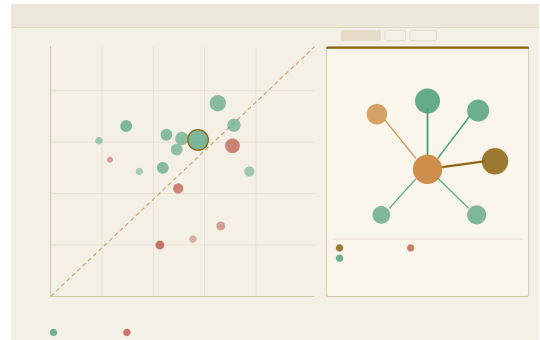
### View 04 — Dynasties

Core MVP · M1 Feedback

The main view is an **original vs sequel scatterplot**: x-axis is the original film's revenue, y-axis is the first sequel's revenue, and the diagonal line represents equal performance. Points above the line improved; points below declined. Color encodes percentage change. Clicking any franchise opens a **radial drill-down** showing all installments as branching nodes from the original, with node size encoding revenue and branch thickness encoding budget.

**Key interactions:** filter by genre, decade, franchise size, hover for film card, click to open radial drill-down, toggle metric between revenue, ROI, and audience rating.

**Tools:** D3.js v7 scatter, radial tree, `belongs_to_collection` field · Lectures 4, 5



## Tools and Lectures

| View                  | Libraries                                         | Lectures |
|-----------------------|---------------------------------------------------|----------|
| Dossier Board         | D3.js v7 force simulation, brush, CSS transitions | 4, 5, 6  |
| The Flow              | D3.js v7, d3-sankey plugin                        | 4, 5     |
| Profitability Matrix  | D3.js v7 heatmap, violin plot                     | 4, 5, 7  |
| Sequels               | D3.js v7 scatter, radial tree layout              | 4, 5     |
| Shared infrastructure | Vanilla JS, CSS custom properties, GitHub Pages   | 1, 2, 3  |

## Work Breakdown

### Core MVP

These pieces are required for the project to be meaningful. The minimum viable product includes the central bubble explorer, the Sankey flow view, and the sequel comparison, since sequel impact was specifically requested in Milestone 1 feedback.

1. Data pipeline: export `movies_lite.json` from `movies.csv` with only browser-necessary fields (title, year, decade, budget, revenue, roi, genres\_list, director, top\_cast, vote\_average, movielens\_avg\_rating, belongs\_to\_collection)
2. Website structure: navigation, shared CineScope dark studio theme, GitHub Pages deployment, persistent genre and decade filter state across views
3. View 01 Dossier Board: force beeswarm with genre cluster and scatter transitions, hover film cards, lasso selection, quadrant annotations
4. View 02 The Flow: Sankey diagram with path highlighting and cross-filtering into View 01
5. View 03 Profitability Matrix: genre-budget heatmap with decade slider and cell drill-down to violin plot
6. View 04 Sequels: original vs sequel scatter with radial franchise drill-down

### Extra Features

These enhance the project but can be removed without losing the core argument.

1. **Parallel Coordinates Plot:** a PCP with axes for budget, runtime, TMDb rating, MovieLens rating, and ROI. Users brush multiple axes simultaneously to filter films. Powerful but complex to read for a general audience; kept as stretch.
2. **Decade morphing animation:** a play button on View 01 that animates the scatter positions as a decade slider advances, Gapminder-style. Highly engaging, relatively straightforward to add on top of the existing force layout.
3. **Persistent film search:** a search bar fixed in the navigation that highlights a named film across all four views simultaneously. Single feature that makes the whole system feel like a real product rather than separate charts.

## Website

The prototype is live at [com-480-data-visualization.github.io/com-480-project-XLB](https://com-480-data-visualization.github.io/com-480-project-XLB) (GitHub Pages, deployed from `/website` on `main`). The current skeleton includes the full CineScope navigation.